AAE 432 Orbital Analysis
3 credits

1 Course Information/Description

The design and operation of spacecraft is a multidisciplinary endeavor covering many aspects of engineering. This course introduces the physics of the orbits and the space environment as well as mission trajectory analysis including constraints, launch windows, ground tracks, some communications and tracking, some constellations, relative motion and contingencies.

2 Learning Resources and Reference Text

Brightspace Learning Management System:

Access the course videos, materials, and resources via Purdue's BS learning management system. Begin with the Start Here tab, which describes how the course Brightspace is organized. We strongly suggest that you explore and become familiar not only with the site navigation but with content and resources available for this course. See the Student Services widget on the campus homepage for resources such as Technology Help, Academic Help, Campus Resources, and Protect Purdue.

Textbook:

None required. Class notes and supplemental handouts will be available on Brightspace. However, the notes and handouts are intended only to support the material presented in the lecture. Good lecture notes will be very important.

A commercial software package (FreeFlyer) developed specifically for astrodynamics applications will be introduced. However, you may also be required to develop your own programs and/or MATLAB scripts.

Assignments

This course has individual assignments. Each week, students in the class receive a problem set of approximately 3-4 problems. The technical problem requires a solution; there are also discussion questions for interpretation of the data that was computed. The assignments typically include additional assessment with the astrodynamics software package FreeFlyer.

3 Course Learning Objectives

Course Goal: Introduction to the multidisciplinary challenges of spacecraft mission analysis. Equations governing orbital motion as well as mathematical models for significant constraints.

Specific Learning Objectives

- Acquire and apply basic technical knowledge about spacecraft trajectory requirements as well as the significant perturbations for an Earth-orbing vehicle.
- Develop intuition about natural spacecraft orbital motion and basic astrodynamics
- Introduction to the space environment and its impact on the orbits
- Key subsystems and the interactions with the orbital requirements

- Introduction to launch and mission operations

Evaluation

Assignments will be submitted and returned via Gradescope. Information will be available to assist and we will answer questions during office hours. Uploads to Gradescope, such as pdfs, require a minimum level of quality so that the document is clear and readable (although it is not necessary that HW is typed). Assignments are due in Gradescope and must be uploaded on the due date as noted in the assignment. Solutions will be available on the course site on Brightspace.

Grades: PS 65% Exams 35%
This class is graded according to a curve. Final grades will be distributed through a comparison among students based on the assignments. After each assignment, mean and median scores will be shared to help track your standing in the course.